THE TOPOLOGICAL SPACE OF A DECENT ALGEBRAIC STACK IS KOLMOGOROV

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Decency of the reduced closed substack supported on the closure of a point. Invoked inside the counted proof (source lines 10528-10530). Allows the reduction to the irreducible case inside a decent closed substack.
- The specialization lifting lemma for smooth morphisms from schemes to stacks. Invoked inside the counted proof (source lines 10549-10552). Lifts a specialization between generic points to a specialization in the presenting scheme.
- The characterization of points through generic points of a smooth presentation. Used inside the counted proof (source lines 10537-10543). Produces a third generic point of the space from a generic point of a component of the presentation.

Source reference. Original lemma lemma-representable-decent: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10531–10531. Navigation pointer only: the counted text refers to the Stacks item lemma-representable-decent, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-specialization: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10553–10553. Navigation pointer only: the counted text refers to the Stacks item lemma-specialization, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged

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Source reference. Original reference topology-definition-generic-point: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2283–2283. Navigation pointer only: the counted text refers to the Stacks item topology-definition-generic-point, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be a decent algebraic stack. Then $|\mathcal{X}|$ is Kolmogorov (see Topology, Definition topology-definition-generic-point).*

Proof. Let $x_1, x_2 \in |\mathcal{X}|$ with $x_1 \rightsquigarrow x_2$ and $x_2 \rightsquigarrow x_1$. We have to show that $x_1 = x_2$. Let $\mathcal{Z} \subset \mathcal{X}$ be the reduced closed substack with $|\mathcal{Z}|$ equal to $\overline{\{x_1\}} = \overline{\{x_2\}}$. By Lemma lemma-representable-decent we see that $\mathcal{Z}$ is decent. After replacing $\mathcal{X}$ by $\mathcal{Z}$ we reduce to the case discussed in the next paragraph.

Assume $|\mathcal{X}|$ is irreducible with generic points x_1 and x_2. Pick an affine scheme U and $u_1, u_2 \in U$ and a smooth morphism $f : U \to \mathcal{X}$ such that $f(u_i) = x_i$. Then we find a third point $u_3 \in U$ which is the generic point of an irreducible component of U whose image $x_3 \in |\mathcal{X}|$ is *also* a generic point of $|\mathcal{X}|$. Namely, we can simply choose u_3 any generic point of an irreducible component passing through u_1 (or u_2 if you like). In the next paragraph we will show that $x_1 = x_3$ and $x_2 = x_3$ which will prove what we want.

By symmetry it suffices to prove that $x_1 = x_3$. Since x_1 is a generic point of $|\mathcal{X}|$ we have a specialization $x_1 \rightsquigarrow x_3$. By Lemma lemma-specialization we can find a specialization $u_1' \rightsquigarrow u_3$ in U (!) mapping to $x_1 \rightsquigarrow x_3$. However, u_3 is the generic point of an irreducible component and hence $u_1' = u_3$ as desired. □